

1. Administrative & Study Identification

Full Study Title: A Phase 1, Randomized, Observer-Blind, Dose-Ranging Study to Evaluate the Safety, Tolerability, and Immunogenicity of Modified RNA Vaccine Candidates Against Pandemic Influenza in Healthy Adults **Short Title/Acronym:** A Study to Learn How Safe, Tolerable and Capable of Producing an Immune Response is, a Modified RNA Vaccine Against Pandemic Influenza **Protocol Number:** C5561001 **NCT Number:** NCT06179446 **Sponsor Name:** Pfizer **Investigational Product:** Pandemic Influenza modRNA (pdmFlu) Vaccine **Phase of Development:** Phase 1 **Medical Monitor:** Pfizer Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and immunogenicity of multiple doses of a nucleoside-modified mRNA (modRNA) based pandemic influenza vaccine candidate in healthy adults. **Secondary Objectives:** To evaluate the effects of the study vaccine for the possible prevention of influenza compared to an active comparator (licensed influenza vaccine) and a placebo.

2.2 Background & Rationale Influenza is a disease that can spread easily from person to person, causing systemic symptoms such as body aches, fever, and cough. As part of broader pandemic preparedness efforts, there is a need to develop rapidly scalable vaccines targeting emerging strains. The vaccine candidate being evaluated encodes the hemagglutinin (HA) antigen of a pandemic influenza strain from the A/H5 subtype, designed to antigenically mirror the avian strain (H5, Clade 2.3.4.4b) currently causing significant disease in birds and mammals, with recent limited cross-over into humans.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (licensed influenza vaccine) and Placebo-controlled **Blinding:** Observer-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 160 participants **Number of Sites:** 3 (Pfizer Clinical Research Unit - New Haven, CT; Research Centers of America - Hollywood, FL; NYU Langone Health - New York, NY) **Estimated Study Start:** 13-Dec-2023 (Actual) **Estimated Primary Completion:** 23-Sep-2025 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy participants as determined by medical history, physical examination (if required), and the clinical judgment of the investigator. 2. Age: Between the ages of 18 to 49 years old or 65 to 84 years old. 3. Willing and able to follow all scheduled visits, treatment plans, laboratory tests, lifestyle changes, and other study procedures, and capable of signing informed consent.

4.2 Exclusion Criteria 1. Does not meet seasonal influenza vaccine requirements per cohort. 2. Standard Phase 1 vaccine contraindications, e.g., severe allergic reactions to any vaccine component, or immunocompromised states.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered as a single shot (dose-ranging evaluation of multiple doses). **Route of Administration:** Intramuscular (shot in the arm). **Duration of Treatment:** Single vaccination with up to 13 months of follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications unassociated with immunosuppression. **Prohibited:** Other investigational vaccines or therapies within standard washout periods.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Physical Examination
Baseline	Day 0	Randomization, Single Dose Vaccination (modRNA, active comparator, or placebo)
Follow-up	Up to 13 Months	Safety Monitoring, Immunogenicity Assessments (Blood draws for antibody responses)
End of Study	Month 13	Final Safety Assessment, Exit Interview

(Note: Exact visit intervals depend on the full clinical protocol).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety, tolerability, and immunogenicity (antibody responses against the H5 influenza virus) of the modRNA vaccine candidate. **Measurement Method:** Adverse event reporting (local and systemic) and research assays analyzing serum immune responses.

7.2 Secondary & Exploratory Endpoints * Comparison of reactogenicity and safety profile against the licensed influenza vaccine and placebo. * Optimal dose determination for the pdmFlu vaccine.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of local and systemic reactions post-vaccination via patient reports/diaries. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal Safety Review Committee (IRC) and External Data Monitoring Committee (EDMC) for regular safety data review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT), Per-Protocol (PP), and Safety Set. **Sample Size Justification:** Adequately powered (N=160) for Phase 1 dose-finding and initial safety/immunogenicity reads. **Handling of Missing Data:** Standard imputation methods per Pfizer Statistical Analysis Plan (SAP).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine monitoring for Phase 1 site compliance. **Audit Trail:** 21 CFR Part 11 compliant audit trails, including eCRF locking and source data verification.

11. Study Identifiers & Governance

Primary ID: C5561001 **Registry Number:** NCT06179446 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn How Safe, Tolerable and Capable of Producing an Immune Response is, a Modified RNA Vaccine Against Pandemic Influenza **Official Regulatory Title:** A Phase 1, Randomized, Observer-Blind, Dose-Ranging Study to Evaluate the Safety, Tolerability, and Immunogenicity of Modified RNA Vaccine Candidates Against Pandemic Influenza in Healthy Adults

12. Clinical Framework (Influenza Specific)

Primary Condition: Pandemic Influenza (A/H5N1 subtype focus) **Diagnosis Threshold:** N/A - Prophylactic Vaccine Study **Medical Methodology:** Nucleoside-modified mRNA (modRNA) vaccine encoding

the hemagglutinin (HA) antigen **Optimization Target:** Humoral immune response (antibody titers) against H5 Clade 2.3.4.4b

13. Study Procedures & Methodology

Management Pathway: Prophylactic Vaccination **Education Protocol (HCP):** Vaccine administration and observer-blind protocol training. **Education Protocol (Patient):** Symptom diary tracking and reporting education. **Quality Audit Mechanism:** Centralized safety and data monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Up to 13 months **Onsite Visit Cadence:** Baseline, and scheduled immunogenicity blood draw visits. **Remote Monitoring Frequency:** Periodic safety check-ins. **Checklist Review Interval:** Post-vaccination reactogenicity diary review.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Investigator Name	_____	03-Apr-2026				
Clinical Operations Lead	Clinical Lead Name	_____	03-Apr-2026				
Lead Statistician	Statistician Name	_____	03-Apr-2026				